

Ex. No 4: Analyze email headers and detect email spoofing using MHA (Mail Header Analyzer)

Description:

Here's a step-by-step guide on how to execute and analyze an email header:

1. Access the Email Header

- **Gmail:** Open the email, click the three dots (More) in the upper right corner, and select "Show original."
- **Outlook:** Open the email, click on "File," then "Properties," and find the "Internet headers" box.
- **Yahoo:** Open the email, click on the three dots (More), and select "View raw message."

2. Copy the Email Header

- Once you access the header, copy all the text that appears. This text contains various metadata fields about the email's journey from the sender to your inbox.

3. Identify Key Header Fields

- **From:** The sender's email address.
- **To:** The recipient's email address.
- **Date:** The date and time the email was sent.
- **Subject:** The email's subject line.
- **Return-Path:** The return address if the email bounces.
- **Received:** A list of servers the email passed through on its way to the recipient. This is usually the most critical part of header analysis.
- **Message-ID:** A unique identifier for the email.
- **SPF (Sender Policy Framework) / DKIM (DomainKeys Identified Mail):** Indicators of email authentication.

4. Analyze the 'Received' Fields

- The Received fields show the path the email took from the sender to the recipient. These fields are listed in reverse order (from the last server to the first).
- Each Received line will show:
 - **The sending server's hostname/IP address.**
 - **The receiving server's hostname/IP address.**
 - **The date and time the email passed through.**

5. Check for IP Addresses and Hostnames

- Use tools like **WHOIS** or online IP lookup services to identify the geographical location and ownership of the IP addresses found in the Received lines.
- Check if any IP addresses are suspicious or if the hostname does not match the expected sending server.

6. Examine the SPF, DKIM, and DMARC Results

- **SPF:** Checks if the sender's IP is authorized to send emails for the domain.
 - **Pass:** Indicates the IP is authorized.
 - **Fail:** The IP is not authorized and could be a sign of spoofing.
- **DKIM:** Verifies that the email content has not been altered in transit.
 - **Pass:** The email has not been tampered with.
 - **Fail:** The email content may have been altered.
- **DMARC:** A policy that uses SPF and DKIM to authenticate the email.
 - **Pass:** Indicates alignment with SPF/DKIM policies.
 - **Fail:** Indicates possible spoofing or tampering.

7. Analyze the 'Message-ID'

- The Message-ID is unique to each email. It usually includes the domain of the sending server.
- Check if the domain in the Message-ID matches the sender's domain. If not, it could be a sign of spoofing.

8. Look for Anomalies

- **Mismatch in Email Domains:** Check if the From domain matches the domains in Return-Path, Message-ID, and other authentication results.
- **Strange IP Addresses:** Be wary of IP addresses that do not match the known sending server or are from unexpected locations.
- **Suspicious Time Stamps:** Check for significant time discrepancies in the Received lines, which could indicate delays or rerouting.

9. Use Online Tools

- Use online email header analysis tools like **MXToolbox** or **Google's G Suite Toolbox** to automatically parse and analyze the email header.

10. Document and Report Findings

- Document all your findings, especially if you detect phishing or spoofing.
- Report suspicious emails to your IT department or email provider for further investigation.

Example of a Header Analysis

Here's a quick example with a sample header:

vbnet

Copy code

Received: from mail.example.com (mail.example.com [192.0.2.1])

by mail.receiver.com with ESMTP id u29si8604336pjs.40.2023.08.10.07.00.16;

Thu, 10 Aug 2023 07:00:16 -0700 (PDT)

Received: by mail.example.com with SMTP id a1mr1243772ywh.51;

Thu, 10 Aug 2023 07:00:15 -0700 (PDT)

Message-ID: <CA+7eu=4pSeXgQ@mail.example.com>

- **Check IP Addresses:** Lookup 192.0.2.1 and ensure it is associated with mail.example.com.
- **Verify SPF/DKIM:** Check if mail.example.com is authorized to send emails for this domain.
- **Compare Time Stamps:** Ensure the times make sense chronologically without unusual delays.

Screen Shots:

Original message

Message ID	<-jBeaMyITNm1zcTUu9o2hg@geopod-ismtpd-21>
Created on:	14 August 2026 at 22:02 (Delivered after 1 second)
From:	Flightradar24 <newsletter@fr24.com>
To:	nikhiljangama18@gmail.com
Subject:	There are fewer airship pilots than astronauts
SPF:	PASS with IP 167.89.98.228 Learn more
DKIM:	'PASS' with domain fr24.com Learn more
DMARC:	'PASS' Learn more

[Download original](#)

[Copy to clipboard](#)



PricingToolsDelivery CenterMonitoringProductsResourcesLogin

SuperToolMX LookupBlacklistsDMARCDiagnosticsEmail HealthDNS LookupAnalyze HeadersAll Tools

Email Header Analyzer

Paste Header:

```
pDX3Elv-2Bso3DWhNMNsV-2FJlCfBgFbPCU1sGY2quncSAPEcpBVDDQCAGdwwPRjYVdj8QcVe61l=
wMT3dhwmX0abJh6XyvjmkMfGXwJGKx370EC0Lc5v6QrC8jTWRjZT-2FJD-2FOXs3gC7nF0D-2Bw=
xKwvZRhHXqDMskibZl8-2BBRkvfeFn2OA-3D-3D" alt="3D" width="3D" height="3D" style="border="3D"0" style="border="3D" height:1px !important; width:1px !important; border-wid=
th:0 !important; margin-top:0 !important; margin-bottom:0 !important; margin-r=
ight:0 !important; margin-left:0 !important; padding-top:0 !important; padding=
-bottom:0 !important; padding-right:0 !important; padding-left:0 !important;"=
/></body>
</html>
--02da8a98a7d9cb21a3b60549f150966f94dcaae624480fc4dde979a091--
```

Analyze Header

ABOUT EMAIL HEADERS

This tool will make email headers human readable by parsing them according to RFC 822. Email headers are present on every email you receive via the Internet and can provide valuable diagnostic information like hop delays, anti-spam results and more. If you need help getting copies of your email headers, [just read this tutorial](#).

Headers Found

Header Name	Header Value
Delivered-To	nikhiljangama18@gmail.com
X-Received	by 2002:a05:6a20:7fa9:b0:3cc:22bb:8911 with SMTP id adf61e73a8af0-3cc71ac37f9mr8429098637.6.1786725133451; Fri, 14 Aug 2026 09:32:13 -0700 (PDT)
ARC-Seal	i=1; a=rsa-sha256; t=1786725133; cv=none; d=google.com; s=arc-20260327; b=Ht0t4VVzTFu6+bREWUSAAmFrJph+b7+U964/E2yE7fogmt7nPousBX/3dTIwulWNeI uwP1g1cehnfPLaTlqmpLGo/G2ledilKODGtSx2YclhAKHKeqMPq7ifvRJIITGlxqbq7H9 pLo8jdddlv0cmMj37GZLwaCOG8GRdjBORsWiDbI38qDQXFeUU95wRYANhY3Fv6DJn0 JTR7CuEgh64b0HVuTAHbZ1kWeDWNB2yKvjbyPn4bOi9cEHKh7Iil83VwcXI8C6x52EpC KNuqEPWBBAGDijMA8mk8BDfyxklrHMH9wblyYWWRFobJdsMq/HvNa/wEDUcnTpg01Znp vV5Q==
ARC-Message-Signature	i=1; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20260327; h=list-unsubscribe-post:list-unsubscribe-to:reply-to:message-id :subject:mime-version:from:date:dkim-signature:dkim-signature; bh=WWJt2AMda1oUmhmyuHcYyJmCcD/aqMLxIHxU4Y4ZsU=; fh=L9xcH35JYkLv1ZiPv9LLOGk412kHsS0Cz3nw0puK30=; b=cwfw/S9G5eqDpFa iGrF1+anGAs6kmwTortro9ILSE8r5x4LzjkokHQdzl++xg2qS+ SCUHpyVJCHW4i/xph09RxxMHChVhIVPkGtvtL+W/F/JygGii0fSQ5Hr+wSRXG1J8+PKJ d02n7xopwo8aV0skqW6enO/rwe5lvmkbKc6OZbqKQCZNSk816G49FmPNvgcp3zfl8w3X NXbjx80pCUiRf2UENpc3Q1VU06vzg7SMQN3Phk9ZI2i5EflxEDBRl840R7JDB9qajXc TrZ253iCdco02Zy5eeuwxm9v8JkuLwKg3FjWYybDyr2n2xk6qD2Uk/4cOzDfbUk7IHf7 06BA==; data=google.com
ARC-Authentication-Results	i=1; mx.google.com; dkim=pass header.i=@fr24.com header.s=s1 header.b=eA3coyZ; dkim=pass header.i=@sendgrid.info header.s=smtapi header.b="Laiicm7"; spf=pass (google.com: domain of bounces+7728411-e93a-nikhiljangama18=gmail.com@post.fr24.com designates 167.89.98.228 as permitted sender) smtp.mailfrom="bounces+7728411-e93a-nikhiljangama18=gmail.com@post.fr24.com"; dmarc=pass (p=QUARANTINE sp=QUARANTINE dis=NONE) header.from=fr24.com
Return-Path	<bounces+7728411-e93a-nikhiljangama18=gmail.com@post.fr24.com>
Received-SPF	pass (google.com: domain of bounces+7728411-e93a-nikhiljangama18=gmail.com@post.fr24.com designates 167.89.98.228 as permitted sender) client-ip=167.89.98.228;
Authentication-Results	mx.google.com; dkim=pass header.i=@fr24.com header.s=s1 header.b=eA3coyZ; dkim=pass header.i=@sendgrid.info header.s=smtapi header.b="Laiicm7"; spf=pass (google.com: domain of bounces+7728411-e93a-nikhiljangama18=gmail.com@post.fr24.com designates 167.89.98.228 as permitted sender) smtp.mailfrom="bounces+7728411-e93a-nikhiljangama18=gmail.com@post.fr24.com"; dmarc=pass (p=QUARANTINE dis=NONE) header.from=fr24.com
DKIM-Signature	v=1; a=rsa-sha256; c=relaxed/relaxed; d=fr24.com; h=content-type:date:from:mime-version:subject:reply-to:to: list-unsubscribe:list-unsubscribe-post:cc:content-type:date:feed-back-id:from:subject:to; s=s1; t=1786725132; bh=WWJt2AMda1oUmhmyuHcYyJmCcD/aqMLxIHxU4Y4ZsU=; b=eA3coyZ5UPt4EeDYlaeBjpAEafzcHN3dafNzJkNaeyRvJiEOw+2WdCrczoc9Z2Uv adMx09c8RssGAVdE7gyr1CsavyMOhOom7IrVvK0PLCkgENM9BillzVm8c/ZxiP88pKqIE GVknoKjht1n6FWge9AELLB1Umsk/IBHrqJOviGbBkBV/WvjfVGUIXBk+OL/knmv7o4pAG hmvAv0ANL5MhoMnQvVMFR6nKxT6mBGztjDV9Pvui3pknoFeDOiLSkO715D093f4Kzrf eAV16AtjKsm6jAFmOfXq4rPgkgZrjnf5oFtyu/F6bZsqJQbXC+0mBp/OK9NxrKA==

Received Header

Delivered-To: nikhiljangama18@gmail.com
Received: by 2002:a17:907:d310:b0:c20:eddb:4981 with SMTP id vgl6csp2346230ejc; Fri, 14 Aug 2026 09:32:13 -0700 (PDT)
X-Received: by 2002:a05:6a20:7fa9:b0:3cc:22bb:8911 with SMTP id adf61e73a8af0-3cc71ac37f9mr8429098637.6.1786725133451; Fri, 14 Aug 2026 09:32:13 -0700 (PDT)
ARC-Seal: i=1; a=rsa-sha256; t=1786725133; cv=none; d=google.com; s=arc-20260327; b=Ht0t4VVzTFu6+bREWUSAAmFrJph+b7+U964/E2yE7fogmt7nPousBX/3dTIwulWNeI uwP1g1cehnfPLaTlqmpLGo/G2ledilKODGtSx2YclhAKHKeqMPq7ifvRJIITGlxqbq7H9 pLo8jdddlv0cmMj37GZLwaCOG8GRdjBORsWiDbI38qDQXFeUU95wRYANhY3Fv6DJn0 JTR7CuEgh64b0HVuTAHbZ1kWeDWNB2yKvjbyPn4bOi9cEHKh7Iil83VwcXI8C6x52EpC KNuqEPWBBAGDijMA8mk8BDfyxklrHMH9wblyYWWRFobJdsMq/HvNa/wEDUcnTpg01Znp vV5Q==
ARC-Message-Signature: i=1; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20260327; h=list-unsubscribe-post:list-unsubscribe-to:reply-to:message-id :subject:mime-version:from:date:dkim-signature:dkim-signature; bh=WWJt2AMda1oUmhmyuHcYyJmCcD/aqMLxIHxU4Y4ZsU=; fh=L9xcH35JYkLv1ZiPv9LLOGk412kHsS0Cz3nw0puK30=; b=cwfw/S9G5eqDpFa iGrF1+anGAs6kmwTortro9ILSE8r5x4LzjkokHQdzl++xg2qS+ SCUHpyVJCHW4i/xph09RxxMHChVhIVPkGtvtL+W/F/JygGii0fSQ5Hr+wSRXG1J8+PKJ d02n7xopwo8aV0skqW6enO/rwe5lvmkbKc6OZbqKQCZNSk816G49FmPNvgcp3zfl8w3X NXbjx80pCUiRf2UENpc3Q1VU06vzg7SMQN3Phk9ZI2i5EflxEDBRl840R7JDB9qajXc TrZ253iCdco02Zy5eeuwxm9v8JkuLwKg3FjWYybDyr2n2xk6qD2Uk/4cOzDfbUk7IHf7 06BA==; data=google.com
ARC-Authentication-Results: i=1; mx.google.com; dkim=pass header.i=@fr24.com header.s=s1 header.b=eA3coyZ; dkim=pass header.i=@sendgrid.info header.s=smtapi header.b="Laiicm7"; spf=pass (google.com: domain of bounces+7728411-e93a-nikhiljangama18=gmail.com@post.fr24.com designates 167.89.98.228 as permitted sender) smtp.mailfrom="bounces+7728411-e93a-nikhiljangama18=gmail.com@post.fr24.com"; dmarc=pass (p=QUARANTINE dis=NONE) header.from=fr24.com
Return-Path: <bounces+7728411-e93a-nikhiljangama18=gmail.com@post.fr24.com>
Received: from o167-89-98-228.outbound-mail.sendgrid.net (o167-89-98-228.outbound-mail.sendgrid.net. [167.89.98.228]) by mx.google.com with ESMTPS id 41be03b00d24 (envelope-from bounces+7728411-e93a-nikhiljangama18=gmail.com@post.fr24.com) (client-ip=167.89.98.228; envelope-id=41be03b00d24)
Received-SPF: pass (google.com: domain of bounces+7728411-e93a-nikhiljangama18=gmail.com@post.fr24.com designates 167.89.98.228 as permitted sender) client-ip=167.89.98.228; envelope-id=41be03b00d24
Authentication-Results: mx.google.com; dkim=pass header.i=@fr24.com header.s=s1 header.b=eA3coyZ; dkim=pass header.i=@sendgrid.info header.s=smtapi header.b="Laiicm7"; spf=pass (google.com: domain of bounces+7728411-e93a-nikhiljangama18=gmail.com@post.fr24.com designates 167.89.98.228 as permitted sender) smtp.mailfrom="bounces+7728411-e93a-nikhiljangama18=gmail.com@post.fr24.com"; dmarc=pass (p=QUARANTINE dis=NONE) header.from=fr24.com
DKIM-Signature: v=1; a=rsa-sha256; c=relaxed/relaxed; d=fr24.com; h=content-type:date:from:mime-version:subject:reply-to:to: list-unsubscribe:list-unsubscribe-post:cc:content-type:date:feed-back-id:from:subject:to; s=s1; t=1786725132; bh=WWJt2AMda1oUmhmyuHcYyJmCcD/aqMLxIHxU4Y4ZsU=; b=eA3coyZ5UPt4EeDYlaeBjpAEafzcHN3dafNzJkNaeyRvJiEOw+2WdCrczoc9Z2Uv adMx09c8RssGAVdE7gyr1CsavyMOhOom7IrVvK0PLCkgENM9BillzVm8c/ZxiP88pKqIE GVknoKjht1n6FWge9AELLB1Umsk/IBHrqJOviGbBkBV/WvjfVGUIXBk+OL/knmv7o4pAG hmvAv0ANL5MhoMnQvVMFR6nKxT6mBGztjDV9Pvui3pknoFeDOiLSkO715D093f4Kzrf eAV16AtjKsm6jAFmOfXq4rPgkgZrjnf5oFtyu/F6bZsqJQbXC+0mBp/OK9NxrKA==
Content-Type: multipart/alternative; boundary="02da8a98a7d9cb21a3b60549f150966f94dcaae6624480fc4dde979a091"
Date: Fri, 14 Aug 2026 16:32:12 +0000 (UTC)
From: Flightradar24 <newsletter@fr24.com>
Mime-Version: 1.0
Subject: There are fewer airship pilots than astronauts
Message-ID: <-jBeaMy1THmIzctUu9o2hg@geopod-ismtpd-21>
Reply-To: newsletter@fr24.com
X-SG-EID: =?us-ascii?Q?u001=2ETILndKxtTPodbnOVpTegDlspJ5NTB8K1r3WqWav2fz4wsPp01iyF+FY=2F?= =?us-ascii?Q?Cet+xlDtwggcah=2FDfGy=2Fd0IHt2dRKILdmbwXRF?= =?us-ascii?Q?u001=2E3RPxwHvfbZy57uFbcgx07Arz4f8=2Fa3+z0BwdJ51d60g5tc5jd0HeruL1?= =?us-ascii?Q?Y8mXdbSn7mqiltyHAREXG6oscc5f1m8C7VbkFN?= =?us-ascii?Q?1C0To: nikhiljangama18@gmail.com
X-Entity-ID: u001.iy+wlS3W+amKFwXUVPQ0nw==
List-Unsubscribe: =?us-ascii?Q?3Chhttps=3A=2F=2Fu7728411=2Ect=2Sendgrid=2Enet=2Flu=2Funsubscrib=3Foc=3Du001=2E5jBph?= =?us-ascii?Q?HG5xXv5ot3u2fC0cHhSUL58Tclz1=
List-Unsubscribe-Post: List-Unsubscribe=One-Click
--02da8a98a7d9cb21a3b60549f150966f94dcaae6624480fc4dde979a091
Content-Transfer-Encoding: quoted-printable